



DEPARTMENT OF COMPUTER ENGINEERING
College of Electrical and Mechanical Engineering
(CEME)

National University of Sciences and
Technology (NUST)



DE-44 CE (B)

Fall 2022

EC-435 Introduction to IOTs

Project Report

Group Members:

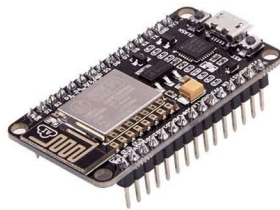
<u>Name</u>	<u>CMS ID</u>
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Hardware:

Hardware components of project is as following and their function which we are getting from them

1. NodeMcu:

NodeMcu is an open source IoT platform that uses the Lua scripting language. It is used to enable IoT applications such as home automation, connected devices, sensing, monitoring, control, and data acquisition. It can be used to connect and interact with other hardware components as well as web services. NodeMcu can be used to create a wide variety of applications, including home automation systems, connected sensors, and data logging.



Use in My project:

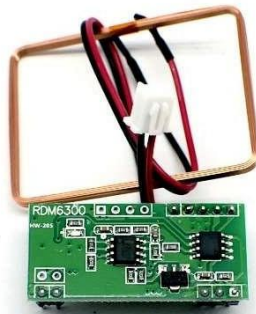
- ☐ *Main Controller of whole project.*
- ☐ *All Components are connected to this.*
- ☐ *It sends the data on localhost through HTTP.*

2. RDM 6300:

The RDM6300 is an RFID reader module based on the EM4100 protocol. It is mainly used for reading and writing RFID tags, and it can be used for applications such as access control, asset management, and security monitoring. Its features include:

- *Supports all EM4100 compatible tags*

- Reads up to 8 bytes of data from the tag
- Writes up to 8 bytes of data to the tag
- Built-in antenna
- Supports serial communication
- Low power consumption
- Compact size

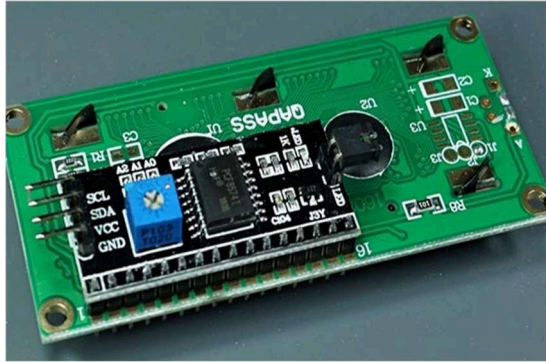


Use in My project:

- ✓ It reads the uid no of card and in bytes and store in temporary variable.
- ✓ The uid or card no are sent on local host by NodeMcu.

3. I2C LCD:

The I2C LCD is an LCD display that uses the I2C communication protocol to interface with an embedded system. It is usually used to display text, numbers, and simple graphics. Its main properties are: -Low power consumption -Easy to use -small size -Low cost -High speed -Flexible addressing -Up to 8-bit data width -Multi-master capability -Supports several bus speeds -Compatible with most microcontrollers. The main function of an I2C LCD is to display text, numbers, and simple graphics. It is used in many applications, such as user interfaces for automation systems, data displays for medical equipment, and information displays for consumer devices.



Use in My project:

✓ It Display different messages and status.

4. LEDs

A Light-Emitting Diode (LED) is a semiconductor device that emits visible light when an electric current passes through it. LEDs have many advantages over traditional incandescent light sources including lower energy consumption, longer lifetime, improved physical robustness, smaller size, and faster switching. LEDs are used in many applications, including indicators, display backlighting, automotive lighting, and general illumination.



Use in My project:

I have use two LEDs.

Green:

It indicates whether attendance is marked on not.

Red:

It indicates is device is connected through internet or not.

5. Jumper wires

Jumper wires are small electrical wires with a connector on one end and stripped wires on the other. They are used to bridge connections, bypass

connections, or make temporary connections between two points on a circuit. They are typically used to make connections between two pieces of equipment or two points on a circuit board. The main properties of jumper wires are their small size, flexibility, and ability to make and break connections quickly and easily.

Use in My project:

✓ *It connects my all components of project with each other.*

6. Box Screws:

✓ *Box is the encloser which enclose all the component and make an attractive look.*

✓ *Screws are used to protect or hold NodeMcu and LCD on their place.*

Pins Configuration:

I2C LCD:

I2C LCD	NodeMcu
VIN	Vin
GND	GND
SCL	D1
SDA	D2

RDM 6300:

RDM6300	NodeMcu
TX	D5
RX	D6
Vin	Vin
GND	GND

LEDs:

LEDs	NodeMcu
Green	D4
Red	D3
GND	GND

Buzzer:

Buzzer	NodeMcu
Vin	D7
GND	GND

Final Hardware:



Software:

The Software used in my project is as following.

1. Arduino IDE

Arduino IDE (Integrated Development Environment) is a free, open-source software used to program Arduino boards. It provides an easy-to-use environment for programming, compiling, and uploading your code to the Arduino board. It includes a text editor for writing code, a message area, a

text console, a toolbar with buttons for common functions and a series of menus. It also provides a library of example sketches that you can use as a starting point for your own code. The Arduino IDE also supports serial communication with the board, allowing you to use the serial monitor for debugging.

Use in My project:

✓ I have done my whole code in Arduino ide Environment my code is as following:

```

/*****libraries*****/
//RFID-----
#include <SPI.h>
#include <SoftwareSerial.h>
//NodeMCU-----
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <LiquidCrystal_I2C.h>
/*****
*****
SoftwareSerial rdm(D5,D6); // Create rdm instance.
LiquidCrystal_I2C lcd (0x27,16,2);
int red_led=D3;
int green_led=D4;
int buzzer=D7;
/*****
/* Set these to your desired credentials. */
const char *ssid = "Badar sial";
const char *password = "safidarsial";
const char* device_token = "37e06568cd0a1fdc";
/*****
String URL = "http://192.168.0.98/rfidattendance/getdata.php"; //computer IP or the server domain
http://localhost/rfidattendance/index.php?login=success
String getData, Link;
String OldCardID = "";
unsigned long previousMillis = 0;
/*****
WiFiClient wifiClient;
void setup() {
  delay(1000);
  Serial.begin(9600);
  SPI.begin(); // Init SPI bus
  rdm.begin(9600);
  lcd.init();          // initialize the lcd
  lcd.clear();
  lcd.backlight();
  pinMode(red_led, OUTPUT);
  pinMode(green_led, OUTPUT);
  pinMode(buzzer, OUTPUT);
  //-----
  connectToWiFi();
}
/*****

```

```

void loop() {
  //check if there's a connection to Wi-Fi or not
  if(!WiFi.isConnected()){
    connectToWiFi(); //Retry to connect to Wi-Fi
  }
  //-----
  if (millis() - previousMillis >= 15000) {
    previousMillis = millis();
    OldCardID="";
  }
  delay(50);

  String CardID = "";
  String c="";
  lcd.clear();
  lcd.setCursor(1,0);
  lcd.print("SCAN YOUR
  RFID"); lcd.setCursor(2,1);
  lcd.print("Thank You");
  delay(1000);
  digitalWrite(red_led, HIGH);
  digitalWrite(green_led, LOW);
  while (rdm.available() > 0) {
    digitalWrite(red_led, LOW);
    delay(5);
    c = rdm.read();
    CardID += c;
    digitalWrite(green_led, HIGH);
    // delay(1000);
  }

  if( CardID == OldCardID ){
    return;
  }
  else{
    OldCardID = CardID;
  }
  SendCardID(CardID);
  delay(1000);
  CardID="";
}

//*****send the Card UID to the website*****
void SendCardID( String Card_uid ){
  Card_uid=Card_uid.substring(1, 11);
  Serial.println("Sending the Card ID");
  if(WiFi.isConnected()){
    HTTPClient http; //Declare object of class HTTPClient
    //GET Data
    getData = "?card_uid=" + String(Card_uid) + "&device_token=" + String(device_token); // Add the Card ID to
the GET array in order to send it
    //GET methode
    Link = URL + getData;
    http.begin( wifiClient,Link); //initiate HTTP request //Specify content-type header

    int httpCode = http.GET(); //Send the request String
    pld = http.getString(); //Get the response pld
  }
}

```

```

Serial.println("link = " + Link); //Print HTTP return code
Serial.println(httpCode);
lcd.print(httpCode); //Print HTTP return code
Serial.println(Card_uid); //Print Card ID
Serial.println(pld);
lcd.print(pld); //Print request response pld

if (httpCode == 200) {
  if (pld.substring(0, 5) == "login") {
    String user_name = pld.substring(5);
    Serial.println(user_name);
    lcd.clear();
    lcd.setCursor(1,0);
    lcd.print(user_name);
    lcd.setCursor(2,1);
    lcd.print("CheckIN");
    digitalWrite(buzzer,HIGH);
    delay(1000);
    digitalWrite(buzzer,LOW);

  }
  else if (pld.substring(0, 6) == "logout") {
    String user_name = pld.substring(6);
    Serial.println(user_name);
    lcd.clear();
    lcd.setCursor(1,0);
    lcd.print(user_name);
    lcd.setCursor(2,1);
    lcd.print("CheckOUT");
    digitalWrite(buzzer,HIGH);
    delay(1000);
    digitalWrite(buzzer,LOW);

  }
  else if (pld == "succesful") {
    lcd.print(pld);

  }
  else if (pld == "available") {
    lcd.print(pld);
  }
  delay(100);
  http.end(); //Close connection
}
}
}
}
//*****connect to the WiFi*****
void connectToWiFi(){
  WiFi.mode(WIFI_OFF); //Prevents reconnection issue (taking too long to connect)
  delay(1000);
  WiFi.mode(WIFI_STA);
  Serial.print("Connecting to ");
  Serial.println(ssid);
  lcd.setCursor(1,0);
  lcd.print("Connecting to");
  lcd.setCursor(2,1);
  lcd.print(ssid);

```

```

WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}
Serial.println("");
Serial.println("Connected");
lcd.setCursor(1,0);
lcd.print(ssid);
lcd.setCursor(2,1);
lcd.print("Connected");

Serial.print("IP address: ");
Serial.println(WiFi.localIP()); //IP address assigned to your ESP

delay(1000);
}
//=====

```

2. Localhost Website

Localhost is a local server environment used for testing and debugging web applications. It is typically used to run web applications from a local machine, such as a computer running the Apache web server. The main function of the localhost website is to enable the developers to test and debug their web applications before deploying them to a live server. It also allows developers to access their web applications on their local machine, rather than having to upload them to a remote server. The main properties of a localhost website are:

- It is hosted on a local machine, not a remote server.
- It is used for testing and debugging purposes.
- It can be used to access web applications without uploading them to a remote server.
 - It is secure and private, as it is not accessible to anyone outside of the local machine.
 - It can be used to host multiple web applications on the same local machine.

Use in My project:

✓ *My all attendance is maintaining on my local host website:*

Final Software:

Login Page:

RFID Attendance

StudentsManage StudentsStudents AttendanceClassesLog In

Please, Login with the Admin E-mail and Password

Badari@gmail.com

...

LOGIN

Forgot your Password? Reset your password

Student page:

StudentsManage StudentsStudents AttendanceClassesEMELog Out

HERE ARE ALL THE STUDENTS

NAME	CMS ID	GENDER	CARD UID	DATE	CLASSES
Saad Kamran	365879	Male	3C00A79CAF	2023-01-16	IOT CLASS
Soban Arif	370262	Male	3C00A8A8F6	2023-01-16	IOT CLASS
Sible Najam	375695	Male	3C00A8AE70	2023-01-16	IOT CLASS
Shaheer Ahmed	367623	Male	3C00A7945E	2023-01-16	IOT CLASS
Ajay Kumar Jaggi	375268	Male	3C00A84261	2023-01-16	IOT CLASS
Paris	386417	Male	3C00A7DDEC	2023-01-16	IOT CLASS
Talimoor	367041	Male	3C00A82642	2023-01-16	IOT CLASS
Muhammad Abdullah	372567	Male	3C00A796F4	2023-01-11	All
Muhammad Ali	370035	Male	3C00A810C3	2023-01-11	IOT CLASS
Badar Hussaini Phabid	370044	Male	3C00A8E000	2023-01-11	IOT CLASS

Managing Student Page:

ADD A NEW STUDENT OR UPDATE HIS INFORMATION OR REMOVE HIM

1 Student Info

Student Name...

CMS ID...

User Email...

2 Additional Info

User Department:

All Departments

☐Female
☒Male

Add Student

CARD UID	NAME	GENDER	CMS ID	DATE	CLASS
3C00A79CAF	Saad Kamran	Male	365879	2023-01-16	IOT CLASS
3C00A8A8F6	Soban Arif	Male	370262	2023-01-16	IOT CLASS
3C00A8AE70	Sibte Najam	Male	375695	2023-01-16	IOT CLASS
3C00A7945E	Shaheer Ahmed	Male	367623	2023-01-16	IOT CLASS
3C00A84261	Ajay Kumar Jaggi	Male	375268	2023-01-16	IOT CLASS
3C00A7DDEC	Paris	Male	386417	2023-01-16	IOT CLASS
✓	None	None	0	2023-01-16	IOT CLASS
3C00A82642	Taimoor	Male	367041	2023-01-16	IOT CLASS
3C00A796F4	Muhammad Abdullah	Male	372567	2023-01-11	All
3C00A810C3	Muhammad Ali	Male	370035	2023-01-11	IOT CLASS
3C00A85822	Badar Hasnain Shehki	Male	398641	2023-01-11	IOT CLASS

Attendance Page:

RFID Attendance

[Students](#)
[Manage Students](#)
[Students Attendance](#)
[Classes](#)
[EME](#)
[Log Out](#)

HERE ARE THE STUDENTS DAILY ATTENDANCE

Log Filter/ Export to Excel

NAME	CMS ID	CARD UID	CLASS	DATE	TIME IN	TIME OUT
------	--------	----------	-------	------	---------	----------

Classes Page:

RFID Attendance

[Students](#)
[Manage Students](#)
[Students Attendance](#)
[Classes](#)
[EME](#)
[Log Out](#)

ADD A NEW CLASS/UPDATE/REMOVE/ENABLE/DISABLE

Your Classes:

New Classes

Instructor Name	Class	Class UID	C.E.D	Mode	C.Config
ATTENDANCE	IOT CLASS	37e06568cd0a1fdc	2023-01-08	Enrollment Attendance	